

Infective Endocarditis: Approach to a Case of Fever and Palpitation

Dr. Muhammad Reihan, Internal Medicine | Source: Fever & Palpitation – Infective endocarditis (31 slides)

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 Bridge from earlier lectures:

- **CAD lecture:** IE is a cause of **acute MR** and **acute AR**, the same "new murmur + pulmonary oedema" picture as papillary muscle rupture after MI.
- **Pericardial diseases: perivalvular abscess** can extend into the conduction system or pericardium.
- **Lung abscess / pulmonary vascular: right-sided (tricuspid) IE in IVDU** throws **septic pulmonary emboli** (multiple cavitating nodules).
- **Rheumatic fever** (the companion lecture) leaves the damaged valves that IE later colonises.

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Cardiac and extracardiac manifestations of IE	Osler vs Janeway vs Roth; FROM JANE
Knowledge	★ Cardiac conditions and risk categories (high / moderate / low) that warrant prophylaxis	"Who gets amoxicillin before a tooth extraction?"
Skill	★ Apply the modified Duke criteria → definite vs possible	Count majors and minors
Skill	Management plan for IE with a complication	Heart block → abscess; HF → surgery

Lecture flow: Opening case → Definition → Etiology (organisms, risk factors, risk classes) → Pathophysiology → Clinical features (constitutional, cardiac, extracardiac, FROM JANE) → Classification (acute / subacute / prosthetic) → Diagnostics (Duke, labs, echo) → Differential → Treatment → Prognosis → Prevention.

2. Main Content (Lecture Order)

2.0 Opening Case (Slides 5-6)

- **Patient:** 72-year-old man wants **medical clearance for a molar extraction**. He feels well and climbs 3 flights of stairs without dyspnoea.
- **History:** HTN, T2DM, IHD. **Aortic valve replacement for severe AS last year**. Two coronary stents 12 years ago.
- **Drugs:** aspirin, **warfarin**, lisinopril, metformin, sitagliptin, simvastatin.
- **Exam:** afebrile. A "grade 6/6" **systolic ejection click** at the right 2nd ICS (= *the prosthetic valve click*).

→ *Most appropriate next step?* (A) **oral amoxicillin 1 h before** · (B) echo before the procedure · (C) avoid nitrous oxide · (D) stop aspirin + warfarin 72 h before · (E) azithromycin 1 h before and 2 h after. (Answered in §7, Q1.)

2.1 Definition (Slide 7)

Infective endocarditis = infection of the **endocardium**, typically affecting **one or more heart valves**.

2.2 Etiology: Organisms (Slides 8-10) ★

Organism	Setting
★ <i>Staphylococcus aureus</i>	Most common cause of ACUTE IE , including people who inject drugs, prosthetic valves, pacemakers/ICDs
★ Viridans streptococci	Most common cause of SUBACUTE IE , especially after dental procedures
<i>Staphylococcus epidermidis</i>	Bacteraemia from infected peripheral venous catheters ([added] and early prosthetic valve IE)
Enterococci (esp. <i>E. faecalis</i>)	IE after nosocomial UTIs ([added] GU/GI instrumentation)
★ <i>Streptococcus gallolyticus</i> subsp. <i>gallolyticus</i> (Sgg; formerly <i>S. bovis</i>)	Associated with colorectal cancer → [added] colonoscopy
HACEK (gram-negative)	Poor dental hygiene / periodontal infection
Fungal (<i>Candida</i> , <i>Aspergillus fumigatus</i>)	Immunosuppressed (HIV, transplant), IVDU, after cardiac surgery, long-term indwelling IV catheters
<i>Coxiella burnetii</i> , <i>Bartonella</i> spp.	Gram-negatives responsible for culture-negative endocarditis

HACEK = **H**aemophilus · **A**ggregatibacter · **C**ardiobacterium · **E**ikenella · **K**ingella (slide 9).

Infective endocarditis: clue in the stem → organism → antibiotic

Clue	Organism	Targeted treatment
Acute, IVDU, prosthetic, pacemaker, tricuspid valve	Staphylococcus aureus	MRSA: vancomycin (MSSA: anti-staph β-lactam)*
Subacute, after DENTAL work, damaged native valve	Viridans streptococci	Penicillin G
Infected peripheral IV line; early prosthetic (≤1 y)	S. epidermidis (CoNS)	Vancomycin + gentamicin + rifampin (prosthetic)
After nosocomial UTI, GU / GI instrumentation	Enterococcus faecalis	Ampicillin + ceftriaxone (or + gentamicin)*
IE + iron-deficiency anaemia, GI bleed → COLONOSCOPY	S. gallolyticus (bovis) colorectal cancer link	Penicillin G / ceftriaxone*
Poor dental hygiene; culture-negative, large vegetations	HACEK oral gram-negative rods	Ceftriaxone (first line)
Culture-negative: farm animals / cats, homeless (lice); immunosupp.	Coxiella · Bartonella · fungi (Candida, Aspergillus)	Serology; fungal → surgery + antifungal*

* added (not on slides). Empiric: native = vancomycin + β-lactam; prosthetic = + gentamicin + rifampin

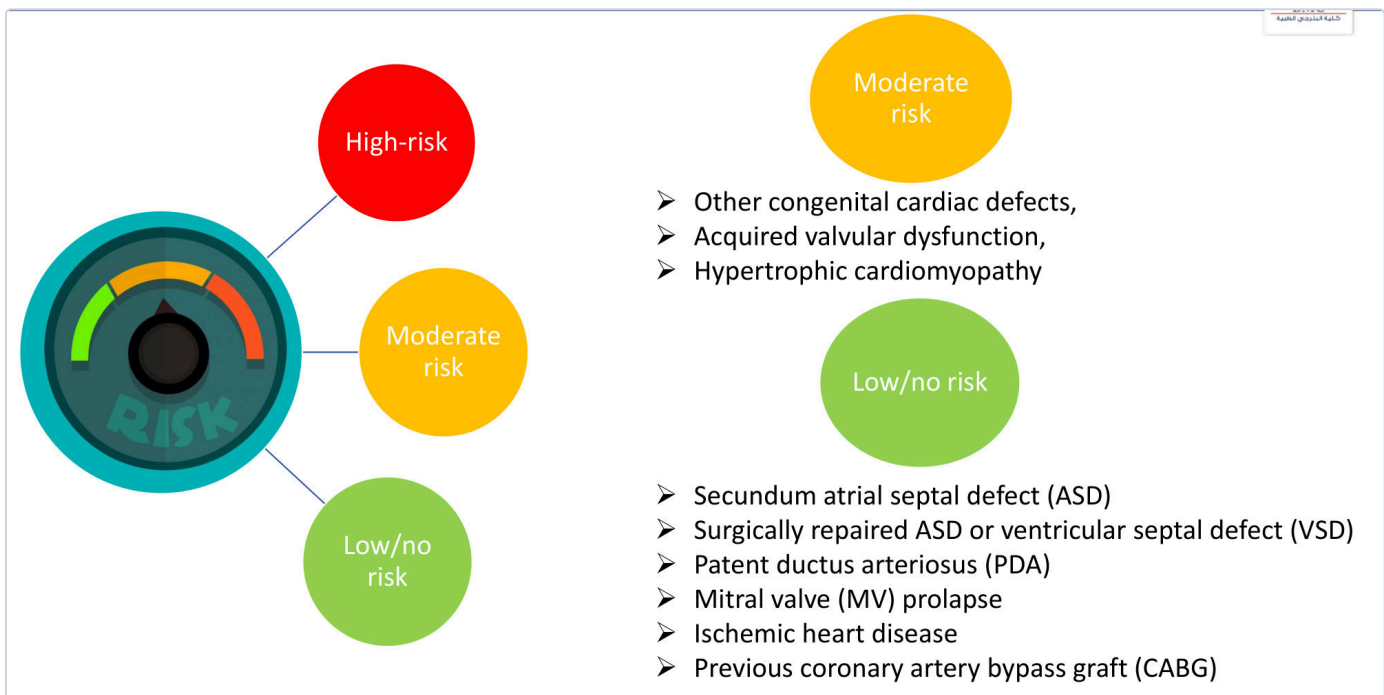
Added diagram: clue → organism → antibiotic

2.3 Risk Factors (Slide 11)

Cardiac	Non-cardiac
Acquired valvular disease	Poor dental status, dental procedures
Prosthetic heart valves	Non-sterile IV injection (IVDU)
Congenital heart defects	Intravascular devices
	Surgery · HIV · haemodialysis

2.4 Risk Classification (Slides 12-13) ★

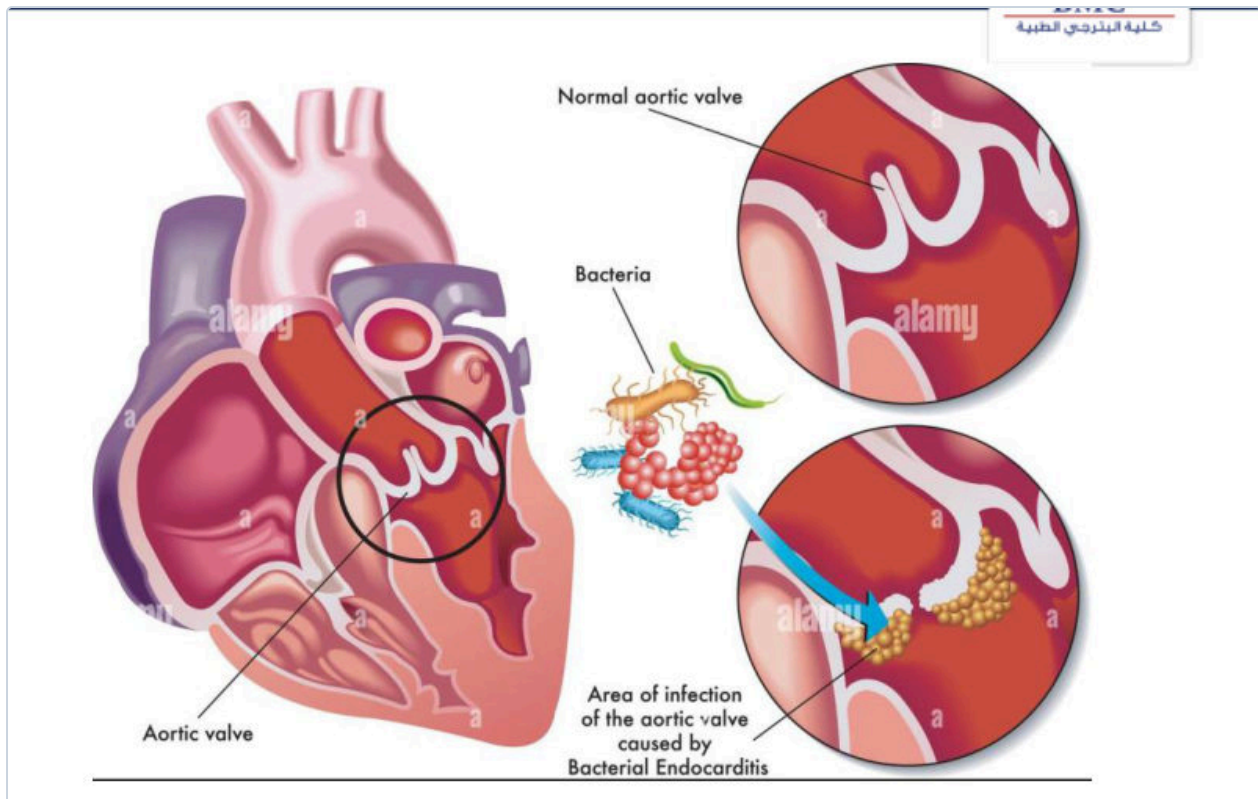
Class	Conditions
● High risk	Prosthetic cardiac valve · previous IE · congenital heart disease (unrepaired, repaired within 6 months , or repaired with residual defects) · cardiac transplant with valve disease · surgically constructed systemic-to-pulmonary shunts or conduits
● Moderate risk	Other congenital cardiac defects · acquired valvular dysfunction (e.g., rheumatic) · hypertrophic cardiomyopathy
● Low / no risk	Secundum ASD · surgically repaired ASD or VSD · PDA (<i>repaired, [added]</i>) · mitral valve prolapse · ischaemic heart disease · previous CABG



Slide 13: high / moderate / low risk categories

2.5 Pathophysiology (Slide 14)

1. **Damaged valvular endothelium** → **sterile vegetation** (platelets + fibrin)
 2. → **Bacterial colonisation** of the vegetation
 3. → **Valve destruction with loss of function** (valve regurgitation)
- **Frequency of valve involvement:** ★ **mitral > aortic > tricuspid > pulmonary.**
 - ★ **Tricuspid valve = most commonly affected in people who inject drugs** (associated with **Pseudomonas**, **S. aureus**, **Candida**).
 - **Clinical consequences:** vegetation → **septic thromboemboli** → vessel occlusion → **infarction**; emboli can seed **metastatic infection** in other organs.



Slide 14: vegetation on the aortic valve

🔄 Recap: Who gets IE and why.

- **Organisms:** *S. aureus* = acute (IVDU, prosthetics, devices); viridans = subacute after dental work; *S. epidermidis* = lines; *Enterococcus* = after UTI; *S. gallolyticus* = colon cancer; HACEK = poor teeth, culture-negative; fungi = immunosuppressed, IVDU; *Coxiella/Bartonella* = culture-negative.
- **Risk classes:** high = prosthetic valve, previous IE, complex/unrepaired CHD, transplant valvulopathy. Moderate = other CHD, acquired valve disease, HCM. Low = ASD, repaired defects, MVP, IHD, CABG.
- **Mechanism:** damaged endothelium → sterile vegetation → colonised → regurgitation + emboli. Mitral is the commonest valve; tricuspid in IVDU.

🔗 **Link:** Vegetations produce three groups of signs: infection (fever), a destroyed valve (murmur) and emboli/immune complexes (peripheral stigmata).

👉 Up next: Clinical features.

2.6 Clinical Features (Slides 15-20) ★

Constitutional: fever and chills, tachycardia, malaise, weakness, weight loss, night sweats, dyspnoea.

Cardiac (slide 16):

- ★ **New murmur, or a change in a pre-existing murmur**
- **Tricuspid regurgitation** in IVDU, immunocompromised patients, congenital heart disease, and **right-heart instrumentation** (central lines)
- **Aortic regurgitation, mitral regurgitation**
- **Heart failure** ([added] the commonest cause of death and the commonest surgical indication)
- ★ **Arrhythmia: a NEW conduction abnormality (e.g., heart block) → suspect a perivalvular ABSCESS**

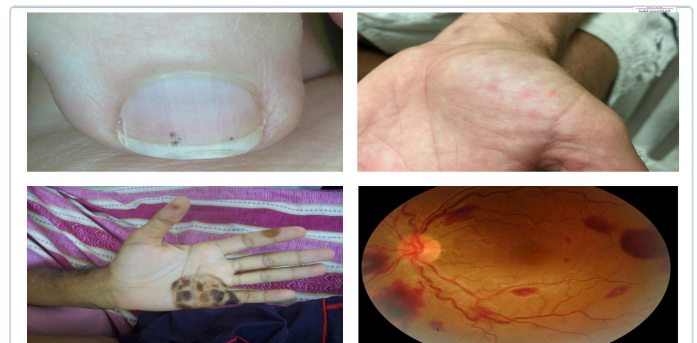
Extracardiac (slides 17-19). Peripheral embolic and immunologic phenomena are seen in **only 5-10%** of patients.

Sign	Description	Mechanism <i>[added]</i>
Petechiae , especially ★ splinter haemorrhages	Under the fingernails	Microemboli / vasculitis
★ Janeway lesions	Small, NON-tender, erythematous macules on palms and soles	Septic emboli (microabscesses)
★ Osler nodes	PAINFUL nodules on the pads of fingers and toes	Immune complex deposition
★ Roth spots	Round retinal haemorrhages with pale centres	Immune complex

Mnemonic: Janeway = **J**ust painless (embolic). **O**sler = **O**uch (immunologic).



Slide 17: splinter haemorrhages, Osler nodes



Slide 18: splinter haemorrhage, Osler node, Janeway lesions, Roth spot

Emboli to other organs (slide 19):

- **Kidney: AKI**, haematuria, anuria (*[added]* emboli or immune-complex GN)
- **Spleen: splenomegaly**, LUQ pain, **splenic abscess**
- **Neurological: ★ septic embolic stroke** (*[added]* also mycotic aneurysm)
- **Pulmonary: septic emboli** (right-sided IE)
- **Others: arthritis**

★ **Mnemonic: FROM JANE** (slide 20) = **F**ever · **R**oth spots · **O**sler nodes · **M**urmur · **J**aneway lesions · **A**cute glomerulonephritis · **N**ail-bed haemorrhage · **E**mbolism.

2.7 Classification (Slide 21) ★

	Acute bacterial (native)	Subacute bacterial (native)	Prosthetic valve IE
Course	Acute onset; fulminant over days-weeks ; severe constitutional symptoms (high fever)	Insidious; weeks-months ; milder	Early: ≤ 1 year after surgery · Late: > 1 year
Pathogens	★ S. aureus (large vegetations that destroy valves); group A strep, <i>S. pneumoniae</i> , <i>N. gonorrhoeae</i>	★ Viridans strep , <i>S. aureus</i> , enterococci, HACEK	Early: CoNS , <i>S. aureus</i> , gram-negative bacilli · Late: CoNS , <i>S. aureus</i> , viridans strep (most common)
Valves	Healthy native valves	Native valves with prior injury or congenital defect	Mechanical and bioprosthetic

	Native valve endocarditis		Prosthetic valve endocarditis
	Acute bacterial endocarditis	Subacute bacterial endocarditis	
Clinical features	• Acute onset • Rapid, fulminant progression (days to weeks) • Severe constitutional symptoms (e.g., high fever)	• Insidious onset • Slow progression (weeks to months) • Less severe constitutional	• Early-onset: ≤ 1 year after surgery • Late-onset: > 1 year after surgery
Main pathogens	• Most common: S. aureus (associated with large vegetations that can destroy the valves) • Others: group A hemolytic streptococci, S. pneumoniae, N. gonorrhoeae	• Viridans streptococci • S. aureus • Enterococci • HACEK	• Early-onset: Coagulase-negative staphylococci, S. aureus, or gram-negative bacilli (most common) • Late-onset: Coagulase-negative staphylococci, S. aureus, Viridans group streptococci (most common)
Affected valves	• Healthy native valves	• Native valves with prior injury or congenital defects	• Mechanical valves • Bioprosthetic valves

Slide 21: classification of IE

📌 **Recap: Clinical picture.** Fever plus a new or changed murmur is IE until proven otherwise. New heart block = perivalvular abscess. Peripheral signs are rare (5–10%): splinters, Janeway (painless, embolic), Osler (painful, immune), Roth spots. Emboli cause stroke, renal infarct/GN, splenic infarct/abscess and septic lung emboli. Acute (*S. aureus*, normal valves, days) vs subacute (viridans, damaged valves, months) vs prosthetic (early CoNS; late resembles native).

🔗 **Link:** Clinical suspicion is converted into a diagnosis with blood cultures + echo, scored by Duke.

👉 **Up next: Diagnostics.**

2.8 Diagnostics (Slides 22–24) ★

- **Suspect IE** from clinical findings (e.g., **fever without a focus + new murmur**) and predisposing conditions.
- The **modified Duke criteria** classify the likelihood as **definite vs possible**.
- ★ **All patients get MULTIPLE blood cultures AND echocardiography.**

Modified Duke criteria (slide 23). Mnemonic: "BE TIMER"

Major: BE	Minor: TIMER
B lood cultures positive: typical organism in > 2 cultures taken 12 h apart	T emperature > 38 °C
E chocardiographic evidence of endocardial involvement (vegetation, abscess, new regurgitation)	I mmunological phenomena (Osler, Roth, GN, [added] RF+)
	M icrobiological evidence (positive culture not meeting a major criterion)
	E mbolic/vascular phenomena (arterial emboli, septic pulmonary emboli, conjunctival haemorrhage, Janeway lesions)
	R isk factors (predisposing heart condition or IV drug use)

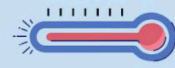
★ Diagnosis	Rule
DEFINITE	2 major · OR 1 major + 3 minor · OR 5 minor
POSSIBLE	1 major + 1 minor · OR 3 minor

Count rule: "2 - 1+3 - 5" definite; "1+1 - 3" possible.

MODIFIED DUKE CRITERIA FOR INFECTIVE ENDOCARDITIS
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MNEMONIC : BE TIMER

MAJOR CRITERIA	MINOR CRITERIA
MNEMONIC - BE	MNEMONIC - TIMER WWW.OPENMED.CO.IN
Blood Culture POSITIVE > 2 times 12 hrs apart. Echocardiographic evidence of Endocardial Involvement 	Temperature > 38 ° C (Fever) Immunological phenomena (Osler's nodes, Roth spots) Microbiological Evidence (Positive blood culture not meeting a major criterion) Embolic Phenomenon (Arterial emboli, septic emboli, conjunctival hemorrhage & painless skin lesions.) Risk Factors - (Congenital heart condition or IV drug use) 

- DEFINITIVE DIAGNOSIS** : Two major criteria OR one major and three minor criteria OR Five minor criteria.
- POSSIBLE DIAGNOSIS**: One major and one minor criteria OR Three minor criteria

Slide 23: modified Duke criteria (BE TIMER)

[added] Blood cultures: take 3 sets from separate sites, ≥ 1 h apart for the first and last, **BEFORE** antibiotics. A single culture of *Coxiella* or phase I IgG > 1:800 counts as a major criterion.

Laboratory (slide 24):

- CBC: leukocytosis with left shift**
- ↑ CRP, ↑ ESR**
- [added] Normocytic anaemia, **haematuria** on urinalysis, ↑ RF

Echocardiography (slide 24):

- ★ **TTE is the INITIAL test of choice** for all suspected IE: vegetations, new regurgitation.
- ★ **TEE if TTE is non-diagnostic** in patients with at least possible IE.
- [added] Go straight to TEE for **prosthetic valves, suspected abscess or intracardiac devices**.

2.9 Differential Diagnosis (Slide 25)

- Non-infective (nonbacterial thrombotic / marantic) endocarditis:** [added] advanced cancer, hypercoagulable states
- ★ **Libman-Sacks endocarditis: verrucous vegetations** in **SLE or antiphospholipid syndrome** ([added] both sides of the leaflet, mitral)
- Prosthetic valve thrombosis**

2.10 Treatment (Slide 26) ★

Setting	Regimen
Empiric: native valve	★ Vancomycin + a β-lactam
Empiric: prosthetic valve	★ Add gentamicin + rifampin
Targeted: MRSA	Vancomycin
Targeted: staphylococcal prosthetic valve IE	Native-valve regimen + gentamicin + rifampin
Targeted: viridans group strep	★ Penicillin G
Targeted: HACEK	★ Ceftriaxone (first line)

[added]:

- Duration:** 4–6 weeks IV.
- MSSA:** cefazolin or an anti-staphylococcal penicillin.
- Enterococcus:** ampicillin + ceftriaxone.
- Surgical indications** (objective: "IE with a complication"): - **Heart failure** from valve dysfunction - **Perivalvular abscess / heart block - Fungal** or highly resistant organism - **Persistent bacteraemia > 5-7 days** on therapy - **Recurrent emboli** with a vegetation > 10 mm - Most **early prosthetic valve IE**
- Do not anticoagulate for the IE itself**, because of haemorrhagic transformation of septic emboli.

2.11 Prognosis (Slide 27)

Adverse prognostic factors: CHF · prosthetic valve infection · valvular/myocardial abscess · embolisation · persistent bacteraemia · altered mental status.

Group	Mortality
Prosthetic valve IE	25-50%
Non-IVDU S. aureus IE	30-45%
IVDU S. aureus or streptococcal IE	10-15% (right-sided disease does better)

2.12 Prevention (Slides 28-29) ★

Who: patients with **high-risk cardiac features** only.

Recommendations for IE prophylaxis (slide 29), high risk:

1. **Prosthetic cardiac valves**
2. **Prosthetic material used for valve repair** (annuloplasty rings, chords)
3. **Previous IE**
4. **Unrepaired cyanotic CHD**, or repaired CHD with **residual shunts or valvular regurgitation** at or next to a prosthetic patch or device
5. **Cardiac transplant with valve regurgitation** from a structurally abnormal valve

Procedures needing prophylaxis (slide 28):

- **Some dental procedures, including tooth extraction** ([added] any manipulation of gingival tissue, the periapical region, or perforation of oral mucosa)
- **Placement of a cardiac implantable electronic device (CIED)**

Regimens: given **30-60 minutes before** the procedure.

- **No penicillin allergy:** ★ **amoxicillin** ([added] 2 g PO single dose)
- **Penicillin allergy:** **macrolide (azithromycin)** or **doxycycline**

[added] Not needed for GI/GU procedures, routine dental cleaning without gingival bleeding, or for MVP/IHD/CABG/stents. Good oral hygiene matters more than antibiotics.

🔄 **Recap: Diagnosis → treatment → prevention.**

- **Diagnosis:** multiple blood cultures + TTE (then TEE). Duke BE TIMER: definite = 2 major, 1 major + 3 minor, or 5 minor; possible = 1 + 1 or 3 minor.
- **Differential:** marantic, Libman-Sacks, valve thrombosis.
- **Empiric treatment:** vanco + β -lactam; prosthetic adds gentamicin + rifampin. **Targeted:** MRSA vanco, viridans penicillin G, HACEK ceftriaxone.
- **Worst prognosis:** prosthetic valve, abscess, HF.
- **Prophylaxis:** only high-risk hearts (prosthetic valve or material, previous IE, complex CHD, transplant valvulopathy), before dental extraction or CIED. Amoxicillin 30-60 min before; azithromycin/doxycycline if allergic.

3. High-Yield Comparison Tables

Table A: Osler vs Janeway vs Roth vs Splinter

Sign	Painful?	Site	Mechanism [added]	Duke category
Osler nodes	✅ Painful	Pads of fingers and toes	Immunologic	Immunologic (minor)
Janeway lesions	❌ Painless	Palms and soles	Embolic microabscess	Vascular (minor)
Roth spots	❌	Retina , pale centre	Immunologic	Immunologic (minor)
Splinter haemorrhages	❌	Nail beds	Embolic / capillary	Nonspecific (trauma also)
Conjunctival haemorrhage	❌	Conjunctiva	Embolic	Vascular (minor)

Table B: Acute vs Subacute vs Prosthetic

	Acute	Subacute	Early PVE	Late PVE
Organism	S. aureus	Viridans	CoNS , S. aureus, GNB	CoNS, S. aureus, viridans
Valve	Normal	Damaged	Prosthetic ≤ 1 y	Prosthetic > 1 y
Speed	Days-weeks	Weeks-months		
Destruction	Severe, large vegetations	Less	Abscess-prone	

Table C: Left- vs Right-Sided IE [added, consistent with slides]

	Left (mitral / aortic)	Right (tricuspid)
Typical patient	Damaged or prosthetic valve, elderly	IVDU , central lines, pacemakers
Organism	Viridans, S. aureus	S. aureus , Pseudomonas, Candida
Emboli to	Brain, kidney, spleen, skin	Lungs (septic emboli, cavitating)
Murmur	MR / AR	TR (louder on inspiration)
Mortality	Higher	Lower (10-15%)

Table D: Infective vs Non-Infective Vegetations

	IE	Libman-Sacks	Marantic (NBTE)
Setting	Bacteraemia + valve damage	SLE / APS	Advanced cancer , hypercoagulable
Cultures	Positive (usually)	Negative	Negative
Vegetation	Destructive, mobile	Verrucous, both leaflet surfaces [added]	Small, sterile, embolise
Fever	✓	±	±

4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Mitral is the commonest valve	IVDU → tricuspid
S. aureus = acute	It is also found in subacute and prosthetic IE
Blood cultures diagnose IE	Culture-negative : prior antibiotics, HACEK , Coxiella , Bartonella , fungi
TTE first	TEE if TTE is non-diagnostic, prosthetic valve or suspected abscess
Peripheral stigmata are classic	They appear in only 5-10%
New murmur + fever → IE	New heart block → perivalvular abscess → surgery
Prophylaxis before dental work	Only high-risk hearts and only for gingival/periapical manipulation or CIED
MVP is a valve lesion	Low/no risk : no prophylaxis (slide 13)
Repaired congenital defect	High risk only within 6 months or with residual defects
Streptococcal IE	S. gallolyticus → colonoscopy

5. Clinical Correlations

- **IVDU with fever, tricuspid murmur louder on inspiration, multiple cavitating lung nodules** → right-sided **S. aureus IE** with septic pulmonary emboli.
- **Elderly patient with subacute fever and weight loss after a dental procedure, MR murmur, splinter haemorrhages** → **viridans** → penicillin G.
- **IE with S. gallolyticus** → **colonoscopy** for colorectal cancer.
- **New PR prolongation / AV block in aortic valve IE** → **aortic root abscess** → TEE + surgery.
- **Sudden hemiparesis in a febrile patient with a murmur** → **septic embolic stroke**. [added]
Avoid thrombolysis.
- **Fever, haematuria, low complement** → immune-complex **GN** (the "A" of FROM JANE).
- **Prosthetic valve 3 months post-op, fever** → **early PVE** (CoNS) → vanco + gentamicin + rifampin; TEE.
- **Culture-negative IE in a farmer** → **Coxiella**. **Homeless with body lice** → **Bartonella quintana**. [added]
- **SLE patient with a mitral vegetation and negative cultures** → **Libman-Sacks**.
- **Patient with a prosthetic valve before tooth extraction** → **amoxicillin 30-60 min before**.
Do not stop warfarin.

6. Rapid Revision (Last-Day)

1. IE = infection of the endocardium, usually the valves.
2. **S. aureus = acute** (IVDU, prosthetic, devices). **Viridans = subacute** (dental). **S. epidermidis** = lines. **Enterococcus** = after UTI. **S. gallolyticus = colon cancer**. **HACEK = poor teeth**. Fungi = immunosuppressed / IVDU. **Coxiella, Bartonella = culture-negative**.
3. **High risk:** prosthetic valve, previous IE, unrepaired / recently repaired / residual-defect CHD, transplant valvulopathy, surgical shunts.
4. **Moderate:** other CHD, acquired valvular dysfunction, HCM. **Low:** secundum ASD, repaired ASD/VSD, PDA, **MVP**, IHD, **CABG**.
5. Endothelial damage → sterile vegetation → colonised → regurgitation + septic emboli.
6. **Mitral > aortic > tricuspid > pulmonary; tricuspid in IVDU** (*S. aureus*, *Pseudomonas*, *Candida*).
7. Fever + **new or changed murmur**. **New heart block = abscess**.
8. Peripheral signs in only **5-10%**: splinters, **Janeway (painless palms/soles)**, **Osler (painful pads)**, **Roth (retina)**.
9. **FROM JANE**.
10. Acute: days, healthy valve, *S. aureus*. Subacute: months, damaged valve, viridans. **PVE early ≤ 1 y** (CoNS) / late > 1 y.
11. **Duke BE TIMER:** major = **B**lood cultures (≥ 2, 12 h apart), **E**cho. Minor = **T**emp > 38, **I**mmunologic, **M**icrobiologic, **E**mbolic, **R**isk factor.
12. **Definite: 2M / 1M + 3m / 5m. Possible: 1M + 1m / 3m.**
13. Labs: leukocytosis with left shift, ↑CRP/ESR. **TTE first → TEE** if non-diagnostic.
14. DDX: **marantic, Libman-Sacks (SLE/APS)**, prosthetic valve thrombosis.
15. Empiric: **native = vancomycin + β-lactam; prosthetic = + gentamicin + rifampin**.
16. Targeted: **MRSA → vancomycin · viridans → penicillin G · HACEK → ceftriaxone**.
17. Bad prognosis: CHF, prosthetic, abscess, emboli, persistent bacteraemia, altered mental status.
PVE 25-50%; non-IVDU *S. aureus* 30-45%; IVDU 10-15%.
18. Prophylaxis: **high-risk only**, for **dental extraction / gingival work** and **CIED placement**.
Amoxicillin 30-60 min before; allergy → **azithromycin or doxycycline**.

7. Exam-Style Questions

Q1 (lecture opening case). A 72-year-old man with an **aortic valve replacement last year**, on aspirin and warfarin, is well and needs a **molar extraction**. What is the most appropriate next step? A. Oral amoxicillin 1 h before · B. Echo before the procedure · C. Avoid nitrous oxide · D. Stop aspirin and warfarin 72 h before · E. Azithromycin 1 h before and 2 h after

Answer: A. *[reasoning added; key not on slides]*

- A **prosthetic valve** is **high risk**, and **tooth extraction** needs prophylaxis (slides 28–29).
- No allergy → **amoxicillin 30–60 min before** as a single dose.
- He is asymptomatic, so echo is not needed (B).
- **Don't stop warfarin** for a dental extraction; bleeding is managed locally (D) *[added]*.
- Azithromycin is only for **penicillin allergy**, and post-procedure doses are not needed (E).
- The "6/6 systolic ejection click" is the normal mechanical valve click.

Q2. A 28-year-old who injects heroin has fever for 5 days, a pansystolic murmur at the left lower sternal border that increases on inspiration, and CXR showing multiple peripheral cavitating nodules. What is the most likely organism? A. Viridans streptococci · B. **Staphylococcus aureus** · C. Enterococcus faecalis · D. S. gallolyticus · E. Coxiella burnetii

Answer: B.

- **Tricuspid IE** in IVDU (TR louder on inspiration) with **septic pulmonary emboli** points to S. aureus, the most common cause of acute IE in IVDU (slides 8, 14).

Q3. A febrile patient with a new MR murmur has 2 positive blood cultures for *S. mitis* taken 12 h apart. TTE shows no vegetation. Temperature is 38.6 °C and he has a history of rheumatic heart disease. How does Duke classify this, and what is the next test? A. Definite; no further imaging · B. Rejected; look for another source · C. Possible; TEE · D. Definite; repeat TTE in 1 week · E. Possible; repeat blood cultures only

Answer: C. Possible IE → TEE. *[reasoning added]*

- **Major:** blood cultures (typical organism, ≥ 2 , 12 h apart).
- **Minor:** fever and a **predisposing heart condition** (RHD).
- That makes **1 major + 2 minor**, which meets "possible" (1 + 1) but not "definite" (needs 1 + 3).
- TTE is non-diagnostic in at least possible IE → **TEE** (slide 24). A vegetation on TEE would add a second major → definite.

Q4. A patient on treatment for aortic valve IE develops a **new first-degree AV block progressing to second-degree block**. What is the most likely complication? A. Septic embolic stroke · B. Drug toxicity · C. **Perivalvular (aortic root) abscess** · D. Libman–Sacks · E. Acute pericarditis

Answer: C.

- A **new conduction abnormality** → **suspect perivalvular abscess** (slide 16).
- *[added]* Next steps: **TEE**, then **surgery**.

Q5. Which painless lesion on the palms and soles is caused by septic microemboli? A. Osler nodes · B. **Janeway lesions** · C. Roth spots · D. Splinter haemorrhages · E. Erythema marginatum

Answer: B (slide 17).

- **Osler** nodes are **painful** finger pads.
- **Roth** spots are retinal.
- **Erythema marginatum** belongs to rheumatic fever.

8. Exam Pitfalls & Traps

- **Osler = painful (pads); Janeway = painless (palms/soles)**. This is the most common swap.
- **MVP, IHD, CABG, stents, repaired ASD/VSD (> 6 months, no residua)** need **no** prophylaxis.
- **Slide 12** garbles the transplant line ("Cardiac transplant with valve disease (surgically constructed systemic-to-pulmonary shunts or conduits)"). These are **two separate** high-risk items: (1) transplant valvulopathy; (2) surgically constructed shunts/conduits (older AHA list).
- **Slide 13 layout:** the text order makes moderate and low look mixed. The diagram is clear: **moderate = other CHD, acquired valve dysfunction, HCM; low = secundum ASD, repaired ASD/VSD, PDA, MVP, IHD, CABG**. This is the older risk-tier list; current AHA guidance gives prophylaxis to high-risk patients only.
- **Prophylaxis timing:** the slide says **30-60 min before** a single dose. "1 h before and 2 h after" (azithromycin) is outdated.
- **Don't stop anticoagulation** for dental extraction. Option D in the case is a trap.
- **Duke counts:** 1 major + **3** minor = definite; 1 + **1** = possible. Students often say 1 + 2 = definite. It isn't.
- **TTE vs TEE:** TTE is the **initial** test; TEE is more sensitive and is used when TTE is non-diagnostic (or for prosthetic valves/abscess *[added]*).
- **S. gallolyticus (bovis) → colonoscopy**. Missing this step is a classic error.
- **Culture-negative IE:** think prior antibiotics first, then HACEK, Coxiella, Bartonella, fungi.
- **Empiric regimen: gentamicin + rifampin** are added for **prosthetic** valves, not native.
- **Mortality numbers:** PVE 25-50% > non-IVDU S. aureus 30-45% > IVDU / strep 10-15%.